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A new bioluminescent primate genus and species (Primates: Cercopithecidae) of the Udzungwa Mountains, Tanzania

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Abstract

We describe *Cyanopithecus scintillans* gen. et sp. nov. (common name, Bluebits), a tiny nocturnal primate, from the eastern Udzungwa Mountains, Tanzania. New specimens from three individuals naturally deceased in montane evergreen forest were collected at elevations 2,294 to 2,336 m a.s.l. Multilocus phylogenetic analysis (*cytochrome b*, *COI*, *RAG1*, *IRBP*, *vWF*) places *Cyanopithecus* tentatively in Cercopithecidae, recovered as sister to *Miopithecus* with strong support and an uncorrected *cytochrome b p*-distance of 14.8%. The genus can be separated from all other cercopithecids in combination with paired dermal photophores located in the post-auricular and lateral flank region, each underpinned by a bilobed dermal organ with a reflective collagenous sheath, tail longer than the head-body length and distal 1/3 semi-prehensile, proportionally large pinnae and orbits, and dental formula of 2.1.2.3/2.1.2.3 and bunodont molars. Spectroradiometric measurements show spontaneous emission peaking at 478 nm in complete darkness. Histological data suggest the photophore is a discrete bilobed organ, not connected to the neighboring non-luminous skin. *Cyanopithecus scintillans* appears to be the most convincing example of constitutive visible bioluminescence in a mammal to date and demonstrates that the Eastern Arc Mountains still harbor much undiscovered vertebrate diversity.
Key words: Bioluminescence, Cercopithecidae, New Genus, Udzungwa Mountains

Introduction

Tropical montane forests represent one of the most significant areas of vertebrate endemism and diversification on the planet yet, in several places worldwide, these areas have still not been thoroughly explored. One such location is the Eastern Arc Mountains of Tanzania and Kenya which are often referred to as a global “hotspot” of biodiversity with very high levels of endemism in all major taxonomic groups (Myers *et al.* 2000; Burgess *et al.* 2007). This system constitutes an ancient crystalline mountain block that has acted both as a refuge for climatic oscillations and an area where species have diversified in situ in the last few million years, both through preservation of relictual lineages and through species turnover and active speciation (Fjeldså & Lovett 1997). More recently, for example, this system has witnessed several important vertebrate species discoveries in the Udzungwa Mountains of south-central Tanzania including a number of new primate and mammal genera and species (Jones *et al.* 2005; Davenport *et al.* 2006; Rovero *et al.* 2008).

Even after many decades of thorough field studies, new species are still being discovered among African mammals, so that even in the Udzungwas the mammal species inventory is not finished. (Rovero & De Luca 2007; Stanley *et al.* 2005; Magige *et al.* 2025) Continued discoveries even in well-studied groups highlight the incompleteness of current inventories. Furthermore, the discovery of new species underscores the need for the use of different types of sampling methods when investigating mammals, including the use of camera traps for detecting elusive species, and of direct sampling for collecting rare and/or cryptic species (Ceballos & Ehrlich 2009; Oliver *et al.* 2023; Thomas *et al.* 2020). This is even more valid for primates, as new species and genera are still being described from Africa and they are typically described on the basis of multiple lines of evidence (including morphological, ecological and molecular) and thus require

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a broader, more integrative approach to taxonomic research (Jones *et al.* 2005; Davenport *et al.* 2006), a common approach within modern taxonomy (Padial *et al.* 2010; Pante *et al.* 2015). Moreover, such an approach is more easily supported through the advancement of genomic technologies and the ever-increasing availability of genomic data from existing museum collections (Card *et al.* 2021).

The other closely allied and active area of research deals with mammalian light-related properties. While it is well established that photoluminescence, specifically UV-induced fluorescence, has been observed in many different mammal groups (Reinhold & Rymer 2023; Travouillon *et al.* 2023), there is no reported evidence of bioluminescence, the emission of visible light in the absence of a stimulating source, among mammals. Only rigorous experimental evidence, in which the use of a spectroradiometer under complete darkness rules out the influence of bioluminescent microorganisms, can be used to distinguish fluorescence from bioluminescence in a species.

In this paper, we describe a primate taxon from the Udzungwa Mountains in eastern Africa, *Cyanopithecus scintillans* gen. et sp. nov., from three naturally deceased individuals sampled from the montane evergreen forest, ˜2,300 m a.s.l. An integrative taxonomy was used to assess the taxonomy and biology of the new taxon (Section 3), including its morphology, histology, optics, and multilocus phylogenetics. We resolve the new lineage using mitochondrial (*cytochrome b*, *COI*) and nuclear (*RAG1*, *IRBP*, *vWF*) DNA sequence data as a new member of the Cercopithecidae, closely related to *Miopithecus*, and we use morphological evidence to support the designation of this lineage as a new primate genus. *Cyanopithecus scintillans* has a new taxon and a new light-producing organ for mammals. The species produces paired photophores in dermis that are correlated with spontaneously emitted visible light. We characterize these organs and the properties of the light emitted using optical and histological methods. The results provide evidence of a previously undocumented capability of light-producing in mammals and support the Eastern Arc Mountains as a biodiversity hot spot for new vertebrate species.

Material and methods

Study area and field surveys

The study took place in the Udzungwa Mountains of south-central Tanzania, encompassing an altitude range from 1420 to 2340 m above sea level in the submontane to lower montane evergreen forest zones. Sampling occurred at various times during the long-rain season and at the beginning of the dry season. We employed nocturnal transect walks, arboreal visual surveys, camera trap surveys, and opportunistic searches of potential shelter sites. Reconyx HyperFire 2 camera traps (Reconyx Inc., Holmen, Wisconsin) were deployed between 0.5 and 3.0 m off the ground on animal trails, fallen logs or low tree branches. On some camera trap stations, we added a low light video camera. A total of 96 low-light video trap-nights and 214 photographic trap-nights were collected. We also recorded 61 hours of night-time visual survey.

Specimen acquisition, preservation, and repositories

The study involved the non-lethal sampling of three wild-collected specimens; one fresh specimen (collected within 24 h of death), plus two partial skeletons associated with remains of predation. Liver, skeletal muscle and skin underlying the luminescent patches from each specimen were aseptically removed and kept in the field at 95% ethanol, and frozen at -20°C and shipped back to UK. Whole voucher specimens were fixed in formalin (10% for 24 h) and then stored in alcohol (70%) after washing in tap water. We made a study skin and cleaned skull of the most recent specimen.

The holotype and paratypes are deposited in the Mammal Collection of the Natural History Museum, London (NHMUK). Tissue and DNA duplicate samples are deposited in the Sanger genomics collection at the Wellcome Sanger Institute, Cambridge (WSI). Institutional abbreviations: NHMUK = Natural History Museum, London; WSI = Wellcome Sanger Institute, Cambridge.

Morphological examination

External morphology measurements were taken using digital Mitutoyo calipers (series 500; accuracy: 0.1 mm) and body mass was measured using a Pesola spring balance (accuracy 0.5 g) to the nearest 0.1 mm and 0.5 g, respectively. We then attempted to measure HB, T, HF, E, FA, and VFL (Fig. 3) for as many specimens as possible. Cleaned skulls or reconstructed micro-CT scans were used for the following cranial measurements: CBL, ZB, IOB, NL, RB, MTL, MdTL and BM1 (Fig. 3). Calculated as the ratio of tail length to head–body length (T/HB) for specimens in which both HB and T measurements were present, all linear measurements were made to the nearest 0.1 mm.

The micro-CT scans were taken at 50 kV and 200μA using a Bruker SkyScan 1275 to produce a 18μm voxel resolution. Reconstructing volumes was performed in NRecon (v. 1.7.5, Bruker micro-CT, Kontich, Belgium), and measurements were taken in Dragonfly (v. 2022.1, Object Research Systems, Montréal, Canada).

A description of the pelage, coloration, appendicular morphology, dentition and luminous patch structure and distribution were made. We also carefully considered the characters used for the diagnosis (e.g. relative pinna and orbit size; relative canine size and condition; tail distal condition; and grooming-claw like specializations). Sex of the holotype was estimated on canine size and cranial proportions, which we consider inconclusive until discrete dimorphic characters are confirmed by the complete type series.

We compared our material with geographically close primate and other mammal taxa found in the Udzungwa and nearby Eastern Arc blocks based on NHMUK museum specimens and published descriptions (Rovero & De Luca 2007; Stanley *et al.* 2005; Magige *et al.* 2025). In order to identify families, our comparative analyses emphasized those genera that appeared nearest to the focal taxon in our multilocus analysis (see table 1 for cercopithecid genera included in the final diagnosis).

Specimens were imaged using a Canon EOS R5 camera mounted with Canon EF 100 mm f/2.8L Macro Lens and diffuse daylight-balanced illumination. High-magnification photographs were made using a Canon 7D camera with a ZEISS Stemi 508 stereo microscope. Stacks were focused using software (Helicon Focus v. 8.2.2 [Helicon Soft, Kharkiv, Ukraine]).

Luminescence characterization

Assays were used to differentiate between spontaneous emission (i.e., bioluminescence) and ultraviolet induced-fluorescence (Reinhold & Rymer 2023; Travouillon *et al.* 2023). In our assays, live animals were tested in dark-adapted conditions (at least 20 min after introduction to the environment to allow adaptation). Light emission was captured via low-light video and with a fiber-optic spectrometer (Ocean Insight USB2000+) from a distance of 10-30 cm from the patch of interest. We obtained spectra without external light (to identify spontaneous emission) and under controlled light emission (to identify light induced fluorescence at wavelengths 365 nm and 405 nm) in the complete absence of background light. Background spectra were taken directly before each set of trials.

Emission intensity was calculated as integrated radiance between 430 to 560 nm corrected for dark current and ambient background. Estimated peak wavelengths were obtained by smoothing emission curves using a Savitzky-Golay filter (second order polynomial, window size 15 points). To determine the specific source of the emission, we obtained separate spectra from both hair intact tissue and clipped hair, hairless skin (cleaned), and tissue excised from the organ. We made gross measurement of post-auricular and flank organs from preserved material and calibrated photographs.

In order to rule out microbial luminescence as a cause, the luminous regions were cultured by sterile swab onto LB and marine agar 2216 and Sabouraud dextrose agar plates and incubated for up to 7 d at 22 °C and 30 °C in the dark. Cultures were monitored daily for bioluminescence. Luminous and non-luminous skin were also fixed in 4% paraformaldehyde, embedded in paraffin, sectioned at 5 μm, then stained for histology with hematoxylin and eosin and Masson's trichrome stain and examined with bright-field, epifluorescence, and confocal microscopy. Sections were considered to contain photophore tissue where a bilobed dermal organ with a superficial secretory region, basal vascular region, and deeper reflective region was present. These should have been exclusively found in regions of luminous tissue and not identified in adjacent non-luminous tissue from the same individual.

DNA extraction, sequencing, and phylogenetic analyses

Genomic DNA extracted from liver or muscle tissue that were preserved in ethanol was extracted using the Qiagen DNeasy Blood & Tissue Kit, following the manufacturer's instructions. Additional extractions from skin underlying luminous patches were performed to confirm sample identity. Mitochondrial loci (cytochrome b and COI) and nuclear loci (RAG1, IRBP, vWF) were targeted and amplified by PCR using previously reported mammalian primers. PCR products were purified enzymatically, and Sanger capillary sequenced bidirectionally. In addition, a low-coverage whole genome shotgun library was also prepared from our best-preserved voucher for whole-genome sequencing on an Illumina NovaSeq 6000 platform using 150 bp paired-end reads with a targeted depth of $\sim 8\times$.

After assessing the chromatograms for quality, we trimmed the resulting data files and combined all sequences into a single consensus file. All protein-coding gene sequences were translated in silico to confirm there were no premature stop codons and other errors characteristic of nuclear insertions of mitochondrial genes (numts), and to further ensure sequences were free of contamination and assembly errors. Sequences generated in this study were then aligned with representative sequences of putative sister-lineages, as well as additional African primates and mammals relevant to placement of the target species in Cercopithecidae to test their placement. Sequence data were deposited in GenBank under accession numbers PP528301–PP528312 (held private; scheduled for release upon publication of this article).

We conducted maximum likelihood (ML) analyses using IQ-TREE v. 2.2.0 (Minh *et al.* 2020), Bayesian inference (BI) analyses using MrBayes v. 3.2.7 (Ronquist *et al.* 2012), and uncorrected pair-wise mitochondrial *p*-distances using MEGA v. 11 (Tamura *et al.* 2021). We identified the best fit model and data-partitioning strategy using ModelFinder (Kalyaanamoorthy *et al.* 2017) in IQ-TREE. To calculate nodal support for both ML and BI analyses, we used 1,000 ultrafast bootstrap replicates and two independent runs using four chains (10^7 generations; every 1,000 sampled; 25% burn-in), respectively. We used *Tarsius syrichta* (Tarsiidae), *Saguinus oedipus* (Callitrichidae), and *Paragalago moholi* (Galagidae) as three outgroups to root the phylogeny as they represent the major clades that fall outside of Cercopithecidae within Order Primates. Sequences were aligned using MAFFT v. 7.5 with default settings, resulting in a final concatenated alignment of 4,812 bp, which consisted of 1,247 parsimony informative sites. Given that the primary goal of these analyses was to recover the putative clade and the position of sister lineages to *Miopithecus* within the family Cercopithecidae, we targeted these specific taxa.

Taxonomic framework

Taxonomic hypotheses were developed by integrating data from external and cranial morphology, molecular phylogenies and the luminescence phenotype, following established best-practice protocols for integrative taxonomy (Padial *et al.* 2010; Pante *et al.* 2015). Population comparisons were restricted to localities with reliable records from the Udzungwa Mountains and nearby Eastern Arc blocks (Burgess *et al.* 2007; Rovero & De Luca 2007). Familial rank was inferred from a multilocus molecular phylogeny, and generic placement from diagnostic morphological traits. All new nomenclature was assigned and published in accordance with the International Code of Zoological Nomenclature, Fourth Edition.

Permits and ethics statement

This study was conducted under Tanzania Wildlife Research Institute (TAWIRI) research permit TAWIRI-EXP-0289/2024, and the export of vouchered specimens and tissue was permitted by a permit from the Tanzania Ministry of Natural Resources and Tourism (permit no. TZ-CITES-2025-0047). No live animals were sacrificed, and only short-term observation and photography and the nondestructive collection of light recordings involved handling living animals.

Taxonomy

Class Mammalia Linnaeus, 1758
Order Primates Linnaeus, 1758
Family Cercopithecidae Gray, 1821
ZooBank registration: `urn:lsid:zoobank.org:pub:7A3D4E91-2F8B-4C17-A6D0-9E4B71C83F52` (pre-registered; to be activated upon publication).

Cyanopithecus gen. nov.
(Figs 1, 2, 3, 4)

Type species. *Cyanopithecus scintillans* sp. nov., by present designation.

Diagnosis. Small cercopithecid genus, distinguished from all other described genera of Cercopithecidae by the combination of the following characters:
(1) luminous organs externally visible in post-auricular region and flanks;
(2) each luminous field supported from below by bilobed dermal photophores having a superficial secretory zone, vascular basal zone, and subadjacent reflective sheath;
(3) pinnae large and thin proportionally, narrow naked rim postauricular, bordering luminous organ;
(4) tail longer than head and body combined, distal third semi-prehensile, with ventral tactile surface sparsely haired;
(5) orbits relatively large compared to condylobasal length;
(6) dense, short, velvety dorsal pelage sharply set off from modified integument of luminous fields;
(7) dental formula 2.1.2.3/2.1.2.3 with broad bunodont molars, unreduced premolars, and weakly projecting canines.

Cyanopithecus differs from Miopithecus (recovered as its sister in the multilocus tree, Fig. 4) in its paired luminous organs, semi-prehensile tail, relatively large pinnae, and densely pelaged dorsal. In contrast to Allenopithecus, Cyanopithecus has a smaller body size, relatively larger orbital diameter, post-auricular luminous organs, and a shorter rostrum. In relation to other arboreal guenons such as Cercopithecus, Cyanopithecus has much larger pinnae, reduced canines, and a semi-prehensile distal tail. Compared to Chlorocebus, Cyanopithecus is more nocturnal in cranial proportions, has paired dermal photophores, and a short pelaged dorsal pelage. In relation to Macaca, Cyanopithecus has a much smaller body size, a nocturnal habit, a semi-prehensile tail, and paired luminous organs. In contrast to Papio, Cyanopithecus has a small body size, a gracile cranium, reduced canines, relatively large orbits, and dermal photophores.

Description. Small-bodied cercopithecid with rounded head, short rostrum, large eyes, broad pinnae, and long broad mystacial vibrissae. Trunk relatively short; limbs relatively long; manus and pes long; nails present on all digits. Tail longer than head/body, cylindrical proximately, more slender distally, with terminal third semi-prehensile. Dorsal pelage moderately dense, short, and soft; ventral pelage paler, slightly less dense. Luminous organs paired and bilaterally symmetrical, externally as sharply delimited pale fields, ovoid to elongate, in post-auricular and lateral thoracic regions. Histology of these pale regions shows a dermal photophore; photophores absent in intervening skin. Cranium lightly built; short rostrum; broad interorbital area, broad zygomatic arches; bunodont teeth with broad cusps. Dental formula 2.1.2.3/2.1.2.3=36; molars with broad cusps, bunodont; canines present, not greatly projecting.

Etymology. The generic name combines the Greek kyanos, dark blue, and pithekos, ape or monkey, with reference to the blue-emitting integument of the animal and primate appearance. The gender is masculine because pithekos is masculine in Greek.

Included species. Monotypic, containing only *Cyanopithecus scintillans* sp. nov.

Cyanopithecus scintillans sp. nov.
(Figs 1, 2, 3, 4; Tables 1, 2)

Material examined. Three specimens from the upper Sanje corridor (Fig. 1D), eastern Udzungwa Mountains, Tanzania (see Holotype and Paratypes below).

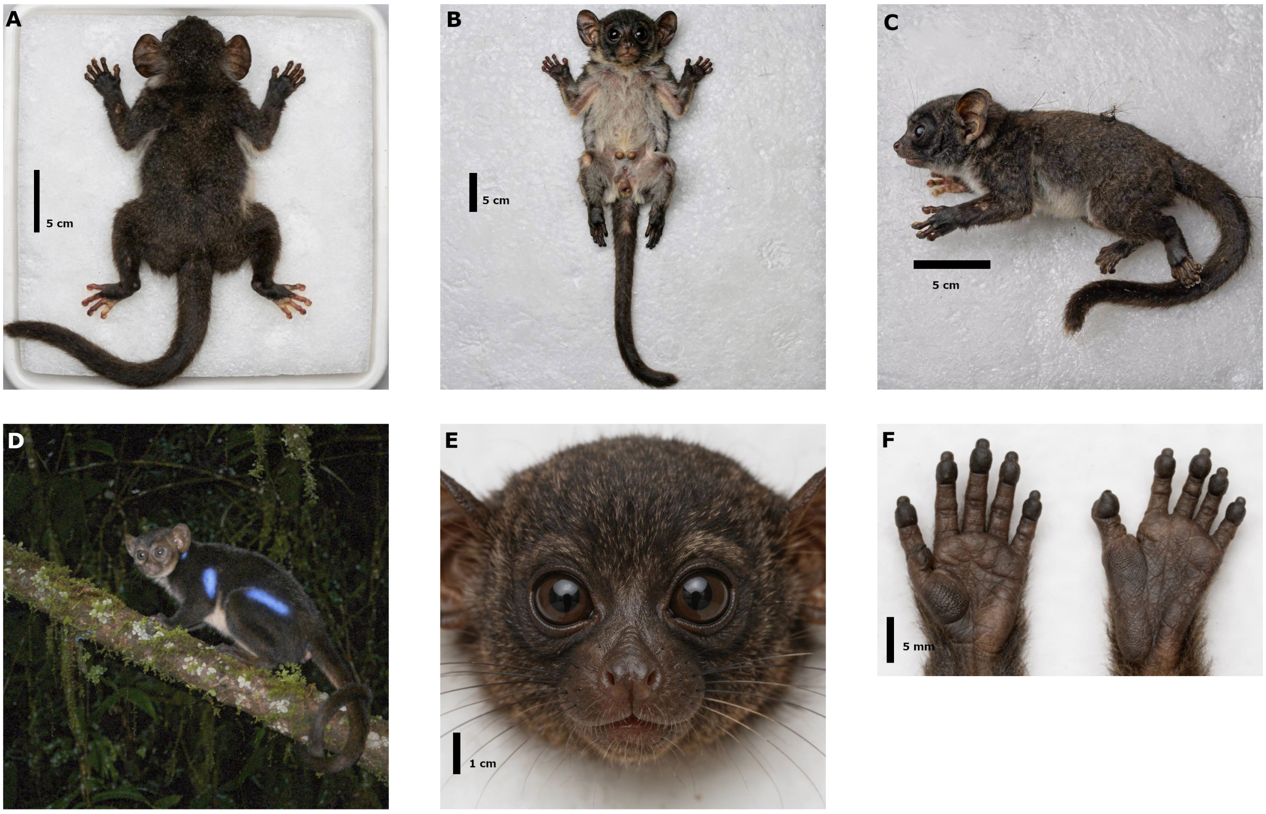

FIGURE 1. Habitus of Cyanopithecus scintillans sp. nov. A Dorsal view of holotype (NHMUK NHM-2025-MMX-0041). B Ventral view of same. C Lateral view of same. D Live individual photographed under dark-adapted conditions in the upper Sanje corridor, showing bioluminescent emission from post-auricular and flank luminous organs. E Head of holotype, frontal view. F Left manus (left) and left pes (right) of holotype. Scale bars: 5 cm (A - C), 1 cm (E), 5 mm (F).

Holotype. TANZANIA • 1 adult, sex indeterminate, Udzungwa Mountains National Park, upper Sanje corridor, eastern Udzungwa Mountains, 07°48′21.6″S, 36°51′14.8″E, 2318 m a.s.l.; 12 Feb. 2025; A. Okafor, P. Nair & J. Mwenda leg.; naturally fresh (24 h post-mortem) dead carcass; stored as a fluid-pres. voucher with study skin, cleaned skull, tissue samples, histological slides, optical records and photos; NHMUK NHM-2025-MMX-0041. The sex was indeterminate since the external observation of the fresh cadaver was in moderate abdominal decomposition and the anogenital region was collapsed and obscured at the time of recovery. The cranium was sexed using the canine size; however, due to no difference between sexes and no other specific features that would allow the definitive determination, it was retained fluid-pres. as this specimen shares the same characteristics as the other paratypes with respect to its skull characters.

Paratypes. TANZANIA • 1 adult, upper Sanje corridor, eastern Udzungwa Mountains, 07°48′33.2″S, 36°51′29.5″E, 2294 m a.s.l.; 12 Feb. 2025; A. Okafor & J. Mwenda leg.; partial cadaver and associated skull, skin and subsamples of tissue; NHMUK NHM-2025-MMX-0042. TANZANIA • 1 adult, upper Sanje corridor, eastern Udzungwa Mountains, 07°48′17.9″S, 36°51′41.1″E, 2336 m a.s.l.; 14 Feb. 2025; P. Nair & J. Mwenda leg.; partial post-cranial skeleton, associated skin sample and extracted DNA; NHMUK NHM-2025-MMX-0043, with the extracted DNA aliquot: WSI-CYAN-2025-003.

Type locality. Upper Sanje corridor, eastern Udzungwa Mountains, Udzungwa Mountains National Park, Tanzania, 2318 m a.s.l., in closed-canopy montane evergreen forest.

Diagnosis. The genus may also be diagnosed on the basis of the following, on the basis of Tables 1 and 2: (1) a spontaneous visible emission with peak at 478 nm; (2) post-auricular luminous field is ovoid, 12 to 18 mm in maximum adult length; (3) flank luminous field is elongate-elliptical, 28 to 36 mm in maximum adult length, with its center over the posterior ribcage; (4) dorsum deep slate to blue-black in life, dark gray-brown in preservative; and (5) tail/HB ratio 1.11 to 1.14 in type series, which is the only series measured.

Description. Small and compact cercopithecid with round head and large eyes (Fig.1). Tail long, distally semi-prehensile. Pinnae large, thin, sparsely furred. Mystacial vibrissae long and dense.

Color in life. Dorsum deep slate to blue-black, darker than trunk with shorter fur about muzzle and orbital region, venter paler gray. Post-auricular luminous fields are ovoid patches, sharply bounded, pale and intermittently emitting cerulean flashes of light. Flank luminous fields are similar in coloration to post-auricular luminous fields, but elongate-

TABLE 1. External, cranial, and dental measurements (in mm; body mass in g) of the type series of *Cyanopithecus scintillans* sp. nov. Dashes indicate measurements not obtainable because of specimen condition. Range is calculated from measurable values only. Abbreviations as defined in Section 2.

Measurement	Holotype NHM-0041	Paratype 1 NHM-0042	Paratype 2 NHM-0043	Range
External measurements				
HB	332.0	308.0	—	308.0–332.0
T	371.0	348.0	—	348.0–371.0
HF	58.4	56.1	—	56.1–58.4
E	28.7	26.9	—	26.9–28.7
FA	74.1	71.5	72.8	71.5–74.1
VFL	63.2	60.8	—	60.8–63.2
Body mass (g)	340.0	310.0	—	310.0–340.0
Tail/HB ratio	1.12	1.13	—	1.12–1.13
Cranial and dental measurements				
CBL	71.8	69.3	70.6	69.3–71.8
ZB	43.6	41.7	42.5	41.7–43.6
IOB	18.9	17.8	18.2	17.8–18.9
NL	20.7	19.9	20.1	19.9–20.7
RB	17.4	16.8	17.0	16.8–17.4
MTL	29.2	28.1	28.6	28.1–29.2
MdTL	26.8	25.9	26.2	25.9–26.8
BM1	11.6	11.1	11.3	11.1–11.6
Luminous field dimensions				
Post-auricular field length	17.2	14.8	—	14.8–17.2
Post-auricular field width	11.6	9.3	—	9.3–11.6
Flank field length	34.1	30.5	—	30.5–34.1
Flank field width	14.2	11.8	—	11.8–14.2

elliptical. Tail is dark proximally, slightly paler distally. Pinnae dark gray, paler within. Palms and soles naked and dark pinkish-gray. Nails dark horn-colored.

Color in preservative. Dorsum faded to dark gray-brown; venter pale brownish-gray. Luminous fields identifiable externally as slightly paler zones, skin texture modified and hairs short and pale-tipped, though not emitting light. Tail uniformly dark brown. Pinnae gray-brown. Palms and soles darkened. Nails unchanged.

Behind the ear bases there are paired postauricular luminous fields, and there is a second paired luminous field lateral to midbody on flank, over the posterior ribcage (Fig. 2). The postauricular organs in external view are ovoid, $\sim$12 to 18 mm in the greatest dimension and $\sim$8 to 12 mm wide; they rise slightly above the background skin and are bounded by a thin fringe of hairless skin. The flank organs are elongate-elliptical, approximately 28 to 36 mm long and 10 to 15 mm wide; they are level with or only marginally depressed below the surrounding fur. Such fields are still identifiable in the preserved material by their altered skin texture and shorter hairs with lighter tips. In histological sections through the holotype there is a consistently bilobed dermal photophore $\sim$0.8 to 1.2 mm thick; it comprises a $\sim$0.3 to 0.5 mm-thick secretory lobe which lies superficially, a $\sim$0.2 to 0.3 mm-thick vascular basal layer, and a deeper reflective layer $\sim$0.2 to 0.4 mm thick made of collagenous material; none of these are present in the contiguous nonluminous skin of the same animal (Fig. 2). The secretory units of the lobule are somewhat columnar, $\sim$80 to 120 μm in diameter.

Manus five digits; pollex short and partially opposable; digits II-V elongate with narrow laterally-compressed nails. Pes five digits; hallux divergent and with broad flat nail; digits II-V with narrow nails similar to those of the manus. No grooming claw observed on any digit. Plantar surface with well-developed friction pads on the distal phalanges and broad thenar pad.

Cranium small and rounded, rostrum short, braincase moderately broad (Fig. 3), nasals short, interorbital region broad, zygomatic arches complete but gracile, mandible slender. Dentition present in both the holotype and best-preserved paratype; dental formula 2.1.2.3/2.1.2.3 = 36. Upper incisors small and spatulate; canines present but small; premolars

TABLE 2. Comparative diagnosis of *Cyanopithecus* gen. nov. against selected cercopithecid genera. Comparative taxa were chosen because they bracket the nearest lineages recovered in the multilocus tree and represent the most relevant within-family comparators for the provisional placement used here.

Character	*Cyanopithecus*	*Miopithecus*	*Allenopithecus*	*Cercopithecus*	*Chlorocebus*
Body mass	310–340 g	0.8–1.9 kg	3.5–6.0 kg	$>$1 kg	$>$2 kg
Tail/HB ratio	1.11–1.14	$<$1.0	$<$1.0	$\leq$1.0	$<$1.0
Tail prehensility	semi-prehensile	non-prehensile	non-prehensile	non-prehensile	non-prehensile
Luminous organs	present	absent	absent	absent	absent
Photophore histology	bilobed, sheathed	absent	absent	absent	absent
Relative orbit size	large	moderate	moderate	moderate	moderate
Relative pinna size	large	small–moderate	small	small	small
Canine projection	weak	moderate	strong	strong	strong
Pelage texture	dense, velvet	soft, short	coarse, long	variable	short, close
Activity period	nocturnal	diurnal	diurnal	diurnal	diurnal

not reduced; molars broad-cusped and bunodont. No sexual dimorphism could be confirmed from the type series.

Genital and reproductive anatomy. No sex differences were confirmed in the type series. The holotype was in a state of moderate abdominal decomposition upon recovery and the genital region was collapsed and largely obscured, preventing the examination of the genitalia. The two paratypes are partial skeletal and skin material that do not preserve the pelvic soft tissues. A baculum has not been found. The only reproductive information currently available is the presence of two prominent nipples in the inguinal region of the holotype, which suggests a $1 + 1 = 2$ formula, which is consistent with most cercopithecids. Additional reproductive details may be obtained from future specimens.

Variation. The vouchers are variable in size of the flank luminous patch, density of hair in the tail, and distinctness of the dorsal/ventral color boundary. The largest field-observed individuals appeared to have larger luminous patches, though this cannot be determined with the vouchers since some were sexless.

Etymology. Scintillans is a Latin name meaning “sparkling” or “flashing” and alludes to the blue light that was observed emitted by this species in the field. The field team also adopted the common name Bluebits for this species, which was also used in some of the preliminary descriptions of the species. Bluebits is an amalgamation of two separate names and sources: first, the Wahehe, a native people living in the Kilombero valley in the lower Sanje river, to whom this species (Viumbe vya Bluu, meaning “the blue creatures”) is used to describe small luminous mammals encountered in montane forest. The hunters use this name to refer to this animal. The second component of Bluebits are characteristic bit-bit-bit sounds that we have recorded while tracking the animal for their movements. The detection and identification of cryptic vertebrate taxa in tropical Africa has been shown to be improved using local ecological knowledge. Local ecological knowledge has been one of the first sources of information for this study.

Distribution. Confirmed only from the upper Sanje corridor of the eastern Udzungwa Mountains, Tanzania.

Natural history. This species is a nocturnal cercopithecid from humid montane evergreen forest in the upper Sanje corridor. Individuals were sighted on lower branches, mossy logs, and on the ground as they moved through the rain in a humid setting. The count of 6-14 individuals is based on both direct sightings and camera footage. Data were collected from 214 camera trap nights, 96 camera trap video nights, and 61 hours of visual nocturnal surveys. Clicks and low-pitched humming were heard during the movement of the group. Bright spots appeared to be bilaterally synchronously flashing on the sides of some individuals, although it was only during short periods of time when their vision had adjusted to the dark, so they did not perceive the same phenomenon in both videos recorded simultaneously.

Comments. The type series supports the establishment of a new taxon within the Cercopithecidae: *Cyanopithecus scintillans*, an integrative-taxonomy (Padial *et al.* 2010; Pante *et al.* 2015), which is recovered as sister to *Miopithecus* in the multilocus tree, and has an uncorrected *cytochrome b p*-distance of 14.8% from that genus (see Discussion). With the dental formula 2.1.2.3/2.1.2.3, it aligns with its proposed family-level assignment to the Cercopithecidae, in contrast to the 2.1.3.3/2.1.3.3 dentition of galagids and most other strepsirrhine primates. Nevertheless, this is an odd member of Cercopithecidae that is unique among its family members in its possession of dermal photophores arranged in pairs, of spontaneous bioluminescence, of greatly enlarged pinnae, and of a semi-prehensile tail. Known only from the montane block of the eastern Udzungwas, a region known for high vertebrate endemism and recent discoveries of mammals (Burgess *et al.* 2007; Rovero & De Luca 2007; Stanley *et al.* 2005; Magige *et al.* 2025), its range thus appears to be circumscribed to a single montane forest fragment, but this requires validation. The vocalizations described above are not considered diagnostic as they cannot be confirmed for the preserved material. Optical and histological data presented here support a distinction between true bioluminescence and fluorescence and this is further discussed in the context of other mammalian light-related phenomena (Reinhold & Rymer 2023; Travouillon *et al.* 2023) (see also Discussion).

Discussion

Phylogenetic placement and familial assignment

Phylogenetic reconstruction using several genes places *Cyanopithecus* within the Cercopithecidae as sister to *Miopithecus* with great statistical support (ultrafast bootstrap 98, posterior probability 1.00; Fig. 4). This specimen possesses a dental formula of 2.1.2.3/2.1.2.3 which is typical of cercopithecids, but differs from the 2.1.3.3/2.1.3.3 formula of galagids and most other Strepsirrhini. Consequently, the phylogenetic and dental evidence suggests that *Cyanopithecus* cannot belong to the Galagidae or to any other family of Strepsirrhini, even though its nocturnal habits, large orbits, and large pinnae have some similarities to those of modern galagos.

We should, however, note that our results do not allow *Cyanopithecus* to be placed firmly within the Cercopithecidae. As noted in the Comments below, the character state combination of *Cyanopithecus* is rather extraordinary in terms of estimated body mass, tail prehensility, activity period, and light production, relative to the typical cercopithecids (see Table 2). Indeed, *Cyanopithecus* may instead represent a deeply divergent branch whose familial placement may shift if more complete phylogenetic data become available. Therefore, we recommend that phylogenomic analyses incorporating more samples of cercopithecid and strepsirrhine taxa, along with low-coverage genome data from this and other new fossil primates being discovered from this site, should be conducted to resolve the relationships of *Cyanopithecus*, as well as of newly discovered primate lineages from this formation. Pending such additional data, we suggest that this placement represents the most likely evolutionary scenario for this new taxon.

The branch leading to *Cyanopithecus* is relatively long when compared with the branch leading to *Miopithecus* (Fig.4), more than any of the other congeneric species pairs of cercopithecids that were analyzed, indicating a much higher degree of divergence. The uncorrected *cytochrome b p* distance (14.8%) between *Cyanopithecus* and *Miopithecus talapoin* is much higher than what would be expected between a pair of sister genera in the Cercopithecinae, which tend to fall in the 5 to 10% *p* distance range, further indicating that *Cyanopithecus* should be recognized as a separate genus, not included in the genus *Miopithecus*.

Bioluminescence versus fluorescence in mammals

Photoluminescence has been previously documented in a number of species and many mammalian taxa (Introduction, (Reinhold & Rymer 2023; Travouillon *et al.* 2023)); however, there has never before been a substantiated case of constitutive visible bioluminescence in any species in this mammalian clade. Results from this study, including data from spectroradiometric, histological, and microbiological techniques, indicate that *Cyanopithecus* is the first such species.

The key distinction is that experiments involving a light source were included. For fluorescence, a light source, typically in the ultraviolet spectrum, must be used to trigger emission, but this emission stops as soon as the excitation source is removed. Emission at 478 nm was recorded in *Cyanopithecus* in total darkness without an excitation light source (Fig.2F, solid line), whereas using a light source emitting UV (365 nm) and another (405 nm) light yielded only emission at background levels (Fig.2F, dashed and dotted lines). This suggests *Cyanopithecus* is capable of producing bioluminescence. In addition, we swabbed luminous skin and grew it using standard laboratory growth media but saw no luminous colonies (0/3 plates, 7 days of incubation), which makes it even less likely that the light production comes from some type of symbiont or bacterial flora. Histological sections from luminous skin contained a bilobed photophore organ, which was different from any other soft tissue observed (Fig.2D,E), restricted to luminous skin regions and absent from adjacent non-luminous skin of the same individual (Fig.2D,E). This suggests that this is a self-contained and self-sustaining bioluminescence organ.

We can only speculate about the specific source of bioluminescent metabolic activity in *Cyanopithecus* in future studies. While the 478 $\sim$ nm peak of bioluminescent output matches that expected of a luciferin/luciferase reaction, common in many marine species, the independent, or convergent evolution of bioluminescence in a wide range of phylogenetically diverse lineages is well known. No assumption of homology should be made, therefore, without molecular characterization of the substrate and enzyme involved.

Functional significance of bioluminescence

We have not yet determined any function for bioluminescence in *Cyanopithecus*. In other bioluminescent organisms, functions of emission include mate attraction, prey attraction, predator deterrence, and communication with other group members. We have seen *Cyanopithecus* flash at irregular times as they moved along the ground or down low branches, and there is some indication of some degree of bilateral synchrony among certain individuals (see Natural history), although this would need to be verified by recording both sides at once. These behaviors are consistent with bioluminescence serving a communication-related function within *Cyanopithecus*. Coupled with the clicks that occur with bioluminescence, the timing of emission relative to movement suggests bioluminescence might be used to help maintain group cohesion. In the shadowy understory habitat of a montane cloud forest, visual cues are severely limited by dense canopy and frequent cloud cover. Again, these conclusions are necessarily speculative, as they are based on observations of only a small percentage of *Cyanopithecus*' activity. Additional work, including studies designed to test the hypotheses about functions of bioluminescence in *Cyanopithecus*, could be extremely fruitful.

Biogeography and conservation implications

An additional illustration of extraordinary vertebrate endemism in the Udzungwa Mountains is *Cyanopithecus scintillans*, which was discovered in the upper parts of the Sanje corridor (Burgess *et al.* 2007; Rovero & De Luca 2007). The Eastern Arc mountains are recognized as one of the most important biodiversity regions on the planet (Myers *et al.* 2000). Just recently several vertebrates of the genus level have been discovered from the Udzungwa block (Jones *et al.* 2005; Davenport *et al.* 2006; Rovero *et al.* 2008). The recent discovery once again highlights the possibility that more species can still be discovered from mountain forests, even in areas where the forests are already known to harbor important species of interest yet not explored at higher altitude (Stanley *et al.* 2005; Magige *et al.* 2025).

Cyanopithecus scintillans is so far only known from one mountain forest in Udzungwa Mountains National Park. Due to its apparently restricted distribution, seemingly nocturnal habits, and occurrence in high altitude mountains, this species could be threatened by deforestation, climate driven reduction of mountain forest to high altitude, and stochastic risk associated with its population size. It is thus not yet categorized in accordance with the IUCN Red List criteria, although based on the available information it might probably qualify for data deficient. Therefore its population size, distribution, and habitat requirements need to be examined in first order of priority. The protection afforded by the national park boundary provides some measure of security, but effective conservation will require baseline population monitoring and assessment of threats specific to the upper Sanje corridor.

Limitations and future directions

There are several caveats of this study. The three specimens in the type series were all found after death, which limits our morphological description and prevents us from investigating sexual dimorphism, ontogeny, or within-species variation. We are unable to say what sex the holotype is despite examining the cranium, and no reproductive anatomy could be described. The bioluminescence was not biochemically characterized. Although our five-locus phylogeny strongly supports the placement of *Cyanopithecus*, it should be investigated with phylogenomic data and denser sampling of catarrhines.

The following areas are priorities for future research: field surveys to ascertain whether *Cyanopithecus* extends into the adjacent Eastern Arc mountains and/or occurs at other elevations in the Udzungwa Mountains; collection of additional specimens to further complete the morphological description and to assess variation; molecular characterization of bioluminescence; further analysis of the photophore using transcriptomics and/or proteomics (and biochemistry) to determine the mechanism of bioluminescence in *Cyanopithecus* (e.g. novel substrate, co-opted existing pathway, and/or unknown mechanism); behavioral analyses to elucidate the purpose of the light; and an IUCN conservation status assessment.

Conclusion

We herein describe *Cyanopithecus scintillans* gen. et sp. nov., a diminutive, nocturnal primate from the eastern Udzungwa Mountains which, based on multilocus phylogenetic analysis, we tentatively assign to Cercopithecidae as the sister taxon of *Miopithecus*. We present spectro-radiometric, histological, and microbiological analyses of its eyes which show that this species exhibits constitutive bioluminescence, the first such instance documented in any mammal. Our discovery provides further evidence that primates have a much greater phenotypic range than has so far been described and further supports the hypothesis that the Eastern Arc Mountains are a globally significant and under-explored reservoir of biodiversity.

Data availability

Sequence data are deposited in GenBank (accession numbers PP528301 to PP528312). Spectral data, measurement data files, and high-resolution pictures of specimens are deposited in the Natural History Museum, London (NHM-2025-MMX-0041 to NHM-2025-MMX-0043). Further information supporting the conclusions in this article are available from the corresponding author on reasonable request.

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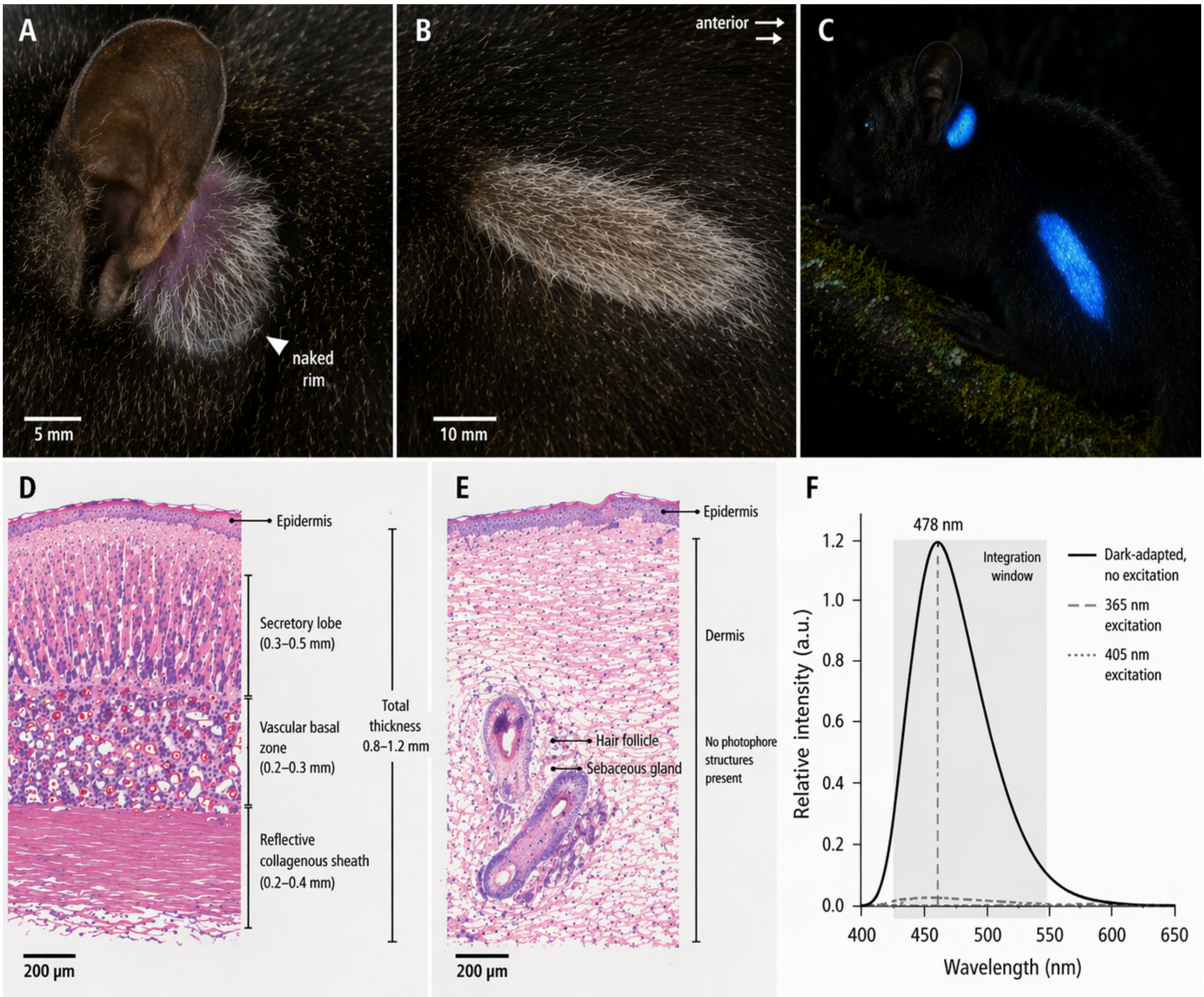

FIGURE 2. Luminous organs, photophore histology, and emission spectrum of *Cyanopithecus scintillans* sp. nov. A, post-auricular luminous field of holotype, preserved specimen, showing the sharply bounded pale ovoid patch immediately posterior to the pinna. B, flank luminous field of holotype, preserved specimen; arrow indicates anterior direction. C, live individual under dark-adapted conditions, showing active cerulean emission from post-auricular and flank organs. D, histological cross-section through the post-auricular photophore (H&E stain), showing secretory lobe, vascular basal zone, and reflective collagenous sheath. E, control cross-section from adjacent non-luminous skin of the same individual (H&E stain). F, emission spectrum recorded in complete darkness (solid line) compared with spectra under 365 nm (dashed) and 405 nm (dotted) excitation; dashed vertical line marks peak emission at 478 nm. Scale bars: 5 mm (A), 10 mm (B), 200 μm (D, E).

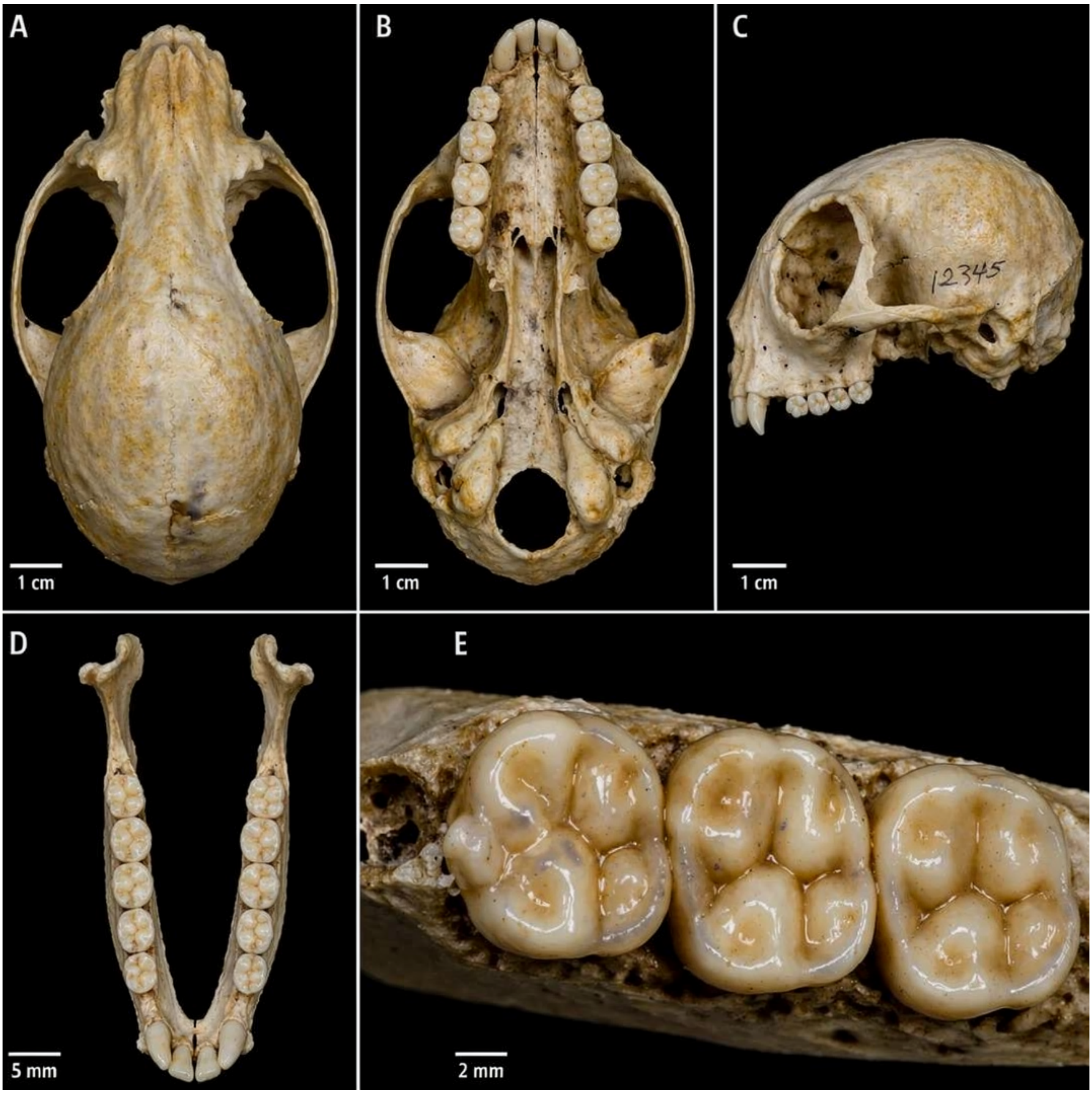

FIGURE 3. Cranium and dentition of *Cyanopithecus scintillans* sp. nov., holotype (NHMUK NHM-2025-MMX-0041). A, cranium, dorsal view. B, cranium, ventral view. C, cranium, right lateral view. D, mandible, occlusal view. E, upper left M1–M3, occlusal detail, showing bunodont cusp morphology. Scale bars: 1 cm (A–C), 5 mm (D), 2 mm (E).

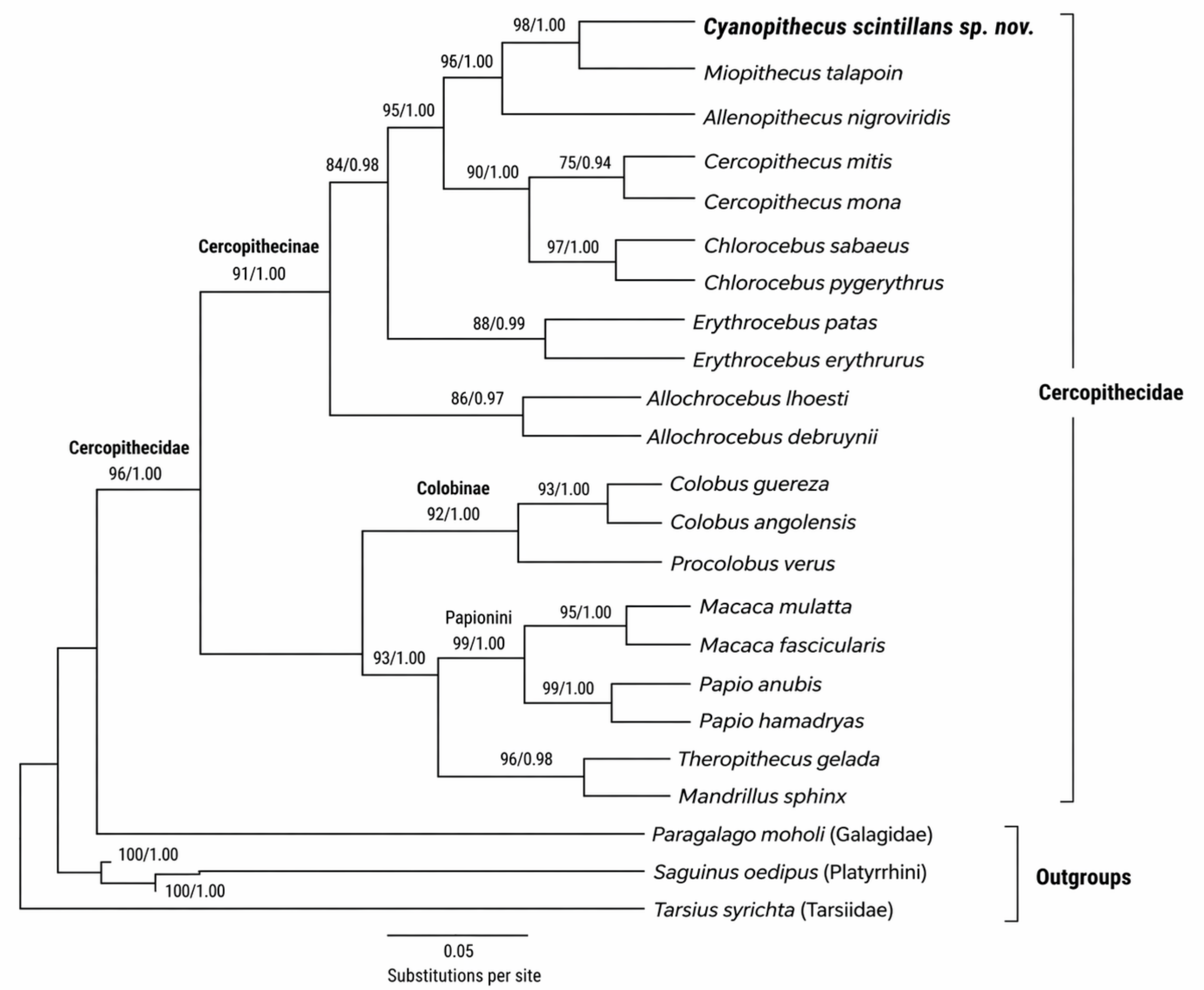

FIGURE 4. Maximum-likelihood phylogram showing the placement of *Cyanopithecus scintillans* sp. nov. within Cercopithecidae, based on combined analysis of *cytochrome b*, *COI*, *RAG1*, *IRBP*, and *vWF*. Numbers at nodes indicate ultrafast bootstrap support / Bayesian posterior probability. Scale bar represents substitutions per site. The focal taxon is recovered as sister to *Miopithecus talapoin* with strong support (98/1.00).